Paleomagnetic dating of burial diagenesis in Mississippian carbonates, Utah

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[1] The objective of this study is to test models for the origin of widespread secondary magnetizations in the Mississippian Deseret Limestone. The Delle Phosphatic Member of the Deseret Limestone is a source rock for hydrocarbons, and modeling studies indicate that it entered the oil window in the Early Cretaceous during the Sevier orogeny. Paleomagnetic and rock magnetic results from the Deseret Limestone and the stratigraphically equivalent Chainman Shale in central and western Utah indicate that the units contain two ancient magnetizations residing in magnetite. Burial temperatures are too low for the magnetizations to be thermoviscous in origin, and they are interpreted to be chemical remanent magnetizations (CRMs). Fold tests from western Utah indicate the presence of a prefolding Triassic to Jurassic CRM. Geochemical ($^{87}Sr/^{86}Sr$, $\delta^{13}C$, and $\delta^{18}O$) and petrographic analyses suggest that externally derived fluids did not alter these rocks. This CRM was acquired at the beginning of the oil window and is interpreted to be the result of burial diagenesis of organic matter. A second younger CRM in western central Utah is apparently postfolding and is probably Late Cretaceous to early Tertiary in age. On the basis of the thermal modeling, the timing overlaps with the oil window. These results are consistent with a connection between organic matter maturation and remagnetization. Modeling of the smectite-to-illite transformation in the Deseret Limestone suggests a mean age prior to acquisition of both CRMs, although the range for illitization overlaps with the Triassic to Jurassic CRM. The results of this study support the hypothesis that pervasive CRMs can be related to burial diagenetic processes. In addition, paleomagnetism can be used to determine the timing of such processes, which can benefit hydrocarbon exploration efforts. *INDEX TERMS:* 1533 Geomagnetism and Paleomagnetism: Remagnetization; 1527 Geomagnetism and Paleomagnetism: Paleomagnetism applied to geologic processes; 1540 Geomagnetism and Paleomagnetism: Rock and mineral magnetism; *KEYWORDS:* remagnetization, burial diagenesis

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1. Introduction

[2] The timing of burial diagenetic processes is essential for reconstructing the history of a formation both stratigraphically and structurally. The timing of these processes is also important for hydrocarbon exploration, i.e., to understand the age relationships between source rock maturation, hydrocarbon migration, and trap formation. Maturation of organic matter is currently dated indirectly by combining data such as conodont alteration, vitrinite reflectance, and burial curves [e.g., *Sandberg and Gutschick*, 1984; *Huntoon et al.*, 1999]. The ability to date organic matter maturation more directly would aid in exploration efforts.

[3] The objective of this study is to test models for the origin of widespread remagnetizations in the Mississippian Deseret Limestone and to specifically test a paleomagnetic method for dating burial diagenetic processes such as maturation of organic matter. Whereas some pervasive chemical remanent magnetizations (CRMs) have been associated with burial diagenetic processes such as organic matter maturation [*Plaster-Kirk et al.*, 1995; *Banerjee et al.*, 1997] or clay diagenesis [e.g., *Katz et al.*, 2000], fluids activated as a result of orogenic activity are also a commonly hypothesized agent of remagnetization [e.g., *McCabe and Elmore*, 1989; *Oliver*, 1992]. Thus the origin(s) of such pervasive CRMs remains unresolved.

[4] The Delle Phosphatic Member of the Deseret Limestone is a source rock for hydrocarbons in other units in Utah [*Huntoon et al.*, 1999]. On the basis of modeling studies, it entered the oil window in the Early Cretaceous during the Sevier orogeny [*Huntoon et al.*, 1999]. To test for a connection between remagnetization and maturation of

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organic matter, samples were collected for fold tests to constrain the timing of any CRMs in the unit. The pole positions were compared to the apparent polar wander path (APWP) to establish the age of the CRMs, which was compared to the depth versus burial curve for the Deseret Limestone [*Huntoon et al.*, 1999] to test for a relationship between the CRMs and maturation of organic matter. Geochemical ($^{87}Sr/^{86}Sr$, $\delta^{13}C$, and $\delta^{18}O$) and petrographic analyses were also conducted to determine if the samples have undergone extensive alteration by orogenic or hydrothermal fluids. The timing of the smectite-to-illite transformation was simulated using the model by *Huang et al.* [1993] to test for a connection between remagnetization and clay diagenesis.

2. Geologic Setting

[5] The Mississippian Deseret Limestone and the laterally equivalent Chainman Shale were deposited in the Deseret Basin of western and central Utah (Figure 1). The Deseret Basin is a foreland basin that formed in response to the Antler Orogeny in the Late Devonian and Early Mississippian in central and eastern Nevada [*Sandberg et al.*, 1980; *Sandberg and Gutschick*, 1984]. The sediments of the Deseret Basin were deposited east of the Antler Highlands at the paleoequator during Mississippian time [*Sandberg and Gutschick*, 1984]. Samples of the Deseret Limestone and Chainman Shale were collected in western and central Utah in the Confusion Range (CR), Mountain Home Range (MHR), Pavant Range (PR), Oquirrh Mountains (OM), Johnson Pass (JNP), and Stansbury Mountains (SM) (Figure 1).

[6] The lithology of the Delle rocks and the lower members of the associated Deseret Limestone and Chainman Shale consists mainly of interbedded claystone, mudstone, shale, micritic limestone, phosphorite, and chert. The rocks generally have a dark gray color, fine texture, and thin bedding [*Sandberg and Gutschick*, 1984]. For this study, the thin micritic limestones, which are interbedded with thick shales, were the primary lithology sampled.

[7] *Sandberg and Gutschick* [1980, 1984] propose that the Osagean to Meramecian Delle Phosphatic Member is the basal member of the Deseret Limestone in central Utah and the Chainman Shale in western Utah. They suggest it was deposited in a regionally extensive, moderately deep (300+ m) basin that was initially sediment-starved but eventually filled with discontinuous encrinite mounds and then phosphatic sediments. *Nichols and Silberling* [1990] do not recognize the Delle Phosphatic Member as a lithologic unit but interpreted the rocks as being deposited during the "Delle phosphatic event." This event is hypothesized to have occurred in a broad, shallow Antler foreland basin in which the Delle rocks were deposited on a shallow portion of the shelf [*Nichols and Silberling*, 1993; *Silberling et al.*, 1997].

[8] The major structural deformation of the Deseret Basin began in the Middle Cretaceous with the Sevier orogeny [*Baer et al.*, 1982; *Nielsen*, 1982]. In Utah, subsidence associated with the onset of the Sevier orogeny began at ~113–91 Ma in Aptian to Cenomanian time [*Heller et al.*, 1986]. The Sevier orogeny resulted in large-scale, eastward propagating thrusts (and associated folds) that extended across western and central Utah. This thrusting ended in the early Tertiary (~50 Ma) and extension began in the

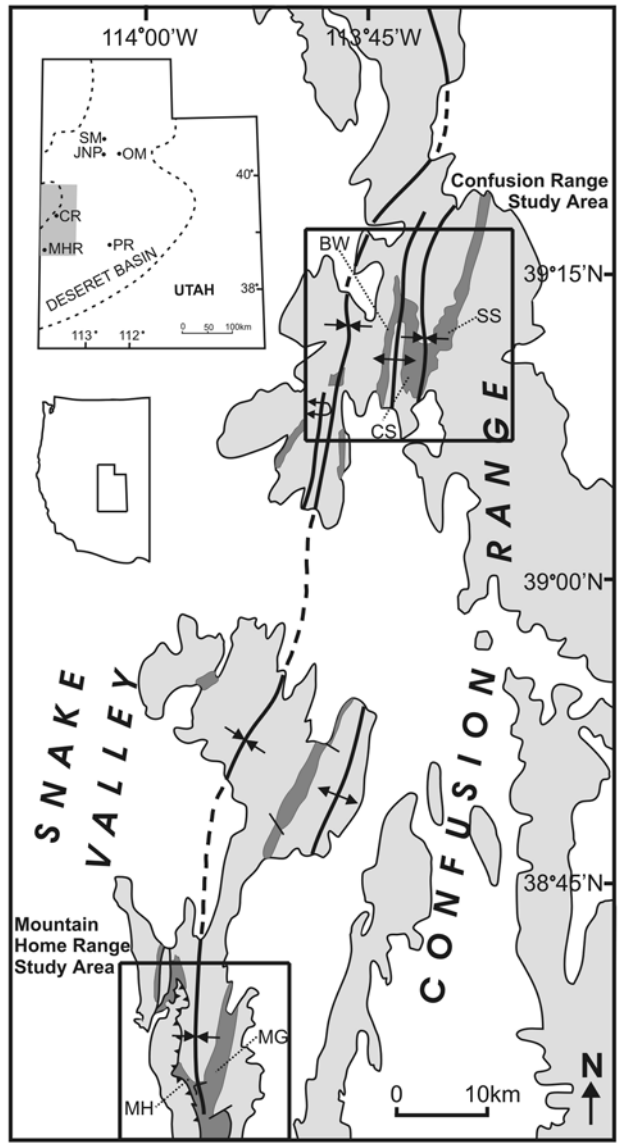


Figure 1. Location of study area in Utah. Small inset map shows the location of Utah in the western United States, and the larger inset map shows the location of the Deseret Basin (latitude 37–44°N, longitude 111–116°W) and the sampling locations in Utah; Confusion Range (CR), Mountain Home Range (MHR), Pavant Range (PR), Stansbury Mountains (SM), Johnson Pass (JNP), Oquirrh Mountains (OM) (modified from *Sandberg and Gutschick* [1984] with permission of RMS/SEPM). The gray rectangle shows the location of the large-scale map. The sampled limbs of the anticline-syncline couplet in the Confusion Range (BW, CS, SS) and the syncline in the Mountain Home Range (MH, MG) are labeled and boxes outline both study areas. The light gray shading represents the mountain ranges and the dark gray shading represents the Chainman Shale.

former Sevier orogenic belt at ~49 Ma [*Bird*, 1998]. The Laramide orogeny (75 to 35 Ma) had little if any effect on the study areas in western Utah [*Bird*, 1998]. In central Utah, structures are dominated by Sevier age thrusts but are superimposed by Laramide uplift and faulting [*Baer et al.*, 1982]. In the late Tertiary and early Quaternary, north-south

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trending normal faults formed the Basin and Range province [*Baer et al.*, 1982]. The Tertiary block faulting either parallels or cuts perpendicular to the Sevier fold and thrust fault axes [*Nielsen*, 1982].

[9] In this study, asymmetric Sevier age folds were sampled. The anticline and syncline sampled in the Confusion Range have steeper west dipping limbs whereas the syncline sampled in the Mountain Home Range has a steeper east dipping limb. Extensional features have also been documented in the Confusion Range that are Eocene to Oligocene in age [*Nichols et al.*, 2002]. *Nichols et al.* [2002] have noted down-to-the-west tilt (25° or less) along the west side of the Confusion Range and believe that there are modest down-to-the-east dips in the Mountain Home Range (N. J. Silberling, personal communication, 2002).

[10] *Huntoon et al.* [1999] have described the Delle Phosphatic Member as the most likely source rock for the Giant Tar Sand Triangle accumulation in eastern Utah. According to the modeling conducted in the Pavant Range by *Huntoon et al.* [1999], the Delle Phosphatic Member reached maturity at ~161 Ma and overmaturity at ~88 Ma. The peak generation in the Pavant Range ($\%R_o$ = 1.0) occurred at ~93 Ma [*Huntoon et al.*, 1999]. The Delle Phosphatic Member contains 3–8 wt % total organic carbon (TOC) of types II and III kerogens in outcrop samples [*Poole and Claypool*, 1984] and higher TOC values (6.7–13.3 wt %) were obtained from core samples [*Morris and Lovering*, 1961]. The lower three members of the Chainman Shale have an average organic carbon content of 1.8 wt % [*Sandberg et al.*, 1980]. The sampling locations in the Confusion Range, Mountain Home Range, and Pavant Range have conodont color alteration index (CAI) values of 1.5 or 2, indicating that these rocks have never been heated to temperatures higher than 90–140°C [*Sandberg et al.*, 1980]. The Oquirrh Mountains have a CAI value of 5 [*Sandberg et al.*, 1980], indicating that these rocks have been heated to temperatures greater than 300°C. Previous paleomagnetic studies of Paleozoic carbonate rocks near the study area have documented the presence of widespread Late Paleozoic magnetizations, usually interpreted as CRMs [e.g., *Gillett and Van Alstine*, 1982; *Gillett and Geissman*, 1993: *Stamatakos et al.*, 1998].

3. Methods

[11] Over 250 samples for paleomagnetic and geochemical study were collected from the Mississippian Deseret Limestone and Chainman Shale at nine locations in western and central Utah (Figure 1). The micritic limestone of the Delle Phosphatic Member was the primary lithology sampled. However, several samples were also taken from overlying members (Needle Siltstone Member, Concretionary Shale Member, Tetro Member, and Uncle Joe Member). In western Utah, the Chainman Shale was sampled on three limbs (locations SS, CS, BW, Figure 1) of an anticline (CS/BW)-syncline (CS/SS) couplet in the Confusion Range. Samples were also collected on both limbs of a syncline (locations MG and MH, Figure 1) in Mountain Home Range ~50 km south of the anticline-syncline couplet. In central Utah, samples were collected in the Oquirrh Mountains, Johnson Pass, and the Pavant Range as well as on a syncline in the Stansbury Mountains (Figure 1).

[12] A gasoline-powered drill was used to core the samples in the field, which were oriented with a compass and inclinometer. In the lab, each core was cut to standard lengths (~2.2 cm) and the natural remanent magnetizations (NRMs) were obtained using a 2G Enterprises cryogenic magnetometer with DC squids located in a magnetically shielded room. The specimens were thermally demagnetized in a magnetically shielded Schonstedt TSD-1 or an ASC thermal demagnetizer oven in 25 steps between the NRM and 700°C. Alternating field (AF) demagnetization of the NRMs up to 120 mT displayed curved demagnetization trajectories and did not clearly resolve the remanent components. As a result, AF treatment was not used to derive directional information.

[13] The demagnetization data were plotted in *Zijderveld* [1967] diagrams, and principal component analysis [*Kirschvink*, 1980] was used to determine magnetic directions. Most specimens had components with mean angular deviations (MAD) of <10° although some had deviations up to 15°. The mean directions were calculated using *Fisher* [1953] statistics and the Super-IAPD99 software. The plunges of the folds were calculated based on measurements made in the field. The paleomagnetic directions (geographic coordinates) were corrected for the plunges prior to application of the fold test (CS/SS fold, N182°E, 2°S; CS/BW fold, N179°E, 4°S; MG/MH fold, N328°E, 18°NW). The *Watson and Enkin* [1993] parameter fold tests were conducted on site means (sites with number of specimens ≥3 (and 4) and $\alpha_{95} \leq 20°$) from individual folds as well as in regional fold tests. Block rotation Fisher analysis [*Enkin and Watson*, 1996] fold tests were also performed to address the possible loss of declination information as a result of block rotations. Specimen directions were used for a fold test on a preliminary data set from the Stansbury Mountains. The fold test results were used to calculate poles that were plotted on the North American APWP constructed by *Van der Voo* [1993].

[14] Low-temperature analysis was performed to assess the possibility that the characteristic remanent magnetization was contaminated by overlap with a resistant thermoviscous or modern viscous remanent magnetization (VRM). After measuring the NRM, representative specimens were submerged in liquid nitrogen (−155°C) and then remeasured. The specimens were warmed to room temperature before undergoing thermal demagnetization.

[15] Isothermal remanent magnetization (IRM) acquisition experiments, as well as AF and thermal demagnetization, were also conducted to determine the magnetic mineralogy of the specimens. The AF demagnetization of the NRM was conducted using a 2G Automated Degaussing System on a set of specimens up to 120 mT and an impulse magnetizer was used to acquire an IRM in 14 steps up to 2500 mT. The specimens underwent AF demagnetization for a second time and then backfield measurements were taken using the impulse magnetizer. Finally, the specimens were subjected to three perpendicular IRMs at 120, 500, and 2500 mT and thermally demagnetized to determine the triaxial IRM decay [*Lowrie*, 1990].

[16] Microscopic analysis was performed on polished thin sections using both reflected and transmitted light to determine the magnetic mineralogy. The rocks were classified using *Dunham*'s [1962] scheme for limestones and diagenetic features were noted.

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[17] The carbon and oxygen isotopic signatures were obtained for 35 micritic limestones from different sampling sites. The samples were prepared by sealed tube reaction of the carbonate with 100% phosphoric acid (H_3PO_4) at 50°C for 2 hours [*McCrea*, 1950; *Swart et al.*, 1991]. The resulting gas was then analyzed for both carbon and oxygen isotope abundances and reported in delta notation (δ) in parts per thousand (‰) relative to the PeeDee Belemnite (PDB) standard. The stable oxygen isotope values were corrected to 25°C after *Swart et al.* [1991].

[18] Strontium isotope analysis was performed on 12 micritic limestone samples at the University of Texas at Austin following the methods described by *Gao et al.* [1992]. Standard SRM 987 was routinely analyzed during this study and yielded 0.710266 ± 0.000007 ($1\sigma = 0.00001$, $n = 3$). The $^{87}Sr/^{86}Sr$ values of the samples were normalized relative to NBS 987 = 0.71014. The $^{87}Sr/^{86}Sr$ values were then compared to the coeval seawater values for Mississippian (Osagean to Meramecian) age limestones as determined by *Denison et al.* [1994].

[19] The timing and extent of the formation of diagenetic illite was simulated using the model proposed by *Huang et al.* [1993]. In this model and other models simulating the smectite to illite transformation [see *Elliott and Matisoff*, 1996], the rate of loss of smectite (C_{sm}) is calculated using an Arrhenius-type expression at each 1 Myr time step from age of deposition to present. The amount of diagenetic illite formed at each time step is the amount of smectite lost at each time step. This model and other models for calculating the amount of illite assumes a nonconserved Al-type reaction where one mole of smectite is consumed for each mole of diagenetic illite formed [*Elliott and Matisoff*, 1996]. The potassium removed from solution to form illite is calculated using a kinetic expression at a given temperature and at a given time (equation (1)). The percentage of potassium removed is a direct proxy for the amount of diagenetic illite formed at each time step:

$$\partial C_{sm}/\partial t = -\kappa (C_{sm})^{\alpha} (K)^{\beta}, \quad (1)$$

where κ is the Arrhenius rate constant. The activation energy is 28 kcal mol^{-1} and preexponential constant is 80,808 s^{-1}. The smectite illitization constant, α, is 2 [*Huang et al.*, 1993]. The burial temperature is calculated at each 1 Myr time step beginning with the age of deposition to present using a constant geothermal gradient (25°C km^{-1}) and the burial curve [*Huntoon et al.*, 1999]. The concentration of potassium (K) is set at 200 ppm. The term β is set equal to 1 in this model and other models [*Huang et al.*, 1993; *Elliott and Matisoff*, 1996]. The time-integrated age of illitization is calculated as the summation of the amount of diagenetic illite formed at each time step and that calculated age of illitization is directly comparable to the measured K-Ar date of illitization [e.g., *Aronson and Hower*, 1976]. For further details of the model, see *Huang et al.* [1993] and *Elliott and Matisoff* [1996].

4. Results and Interpretations

4.1. Paleomagnetism

[20] Mean NRM intensities are 0.511 mA m^{-1} (standard deviation of 0.43, $n = 29$ site means). Thermal demagnetization of specimens resulted in the identification of three components (Figure 2a). A component with northerly declinations and steep down inclinations is removed at low temperatures in most specimens (Figure 2a) and is interpreted as a modern VRM. Component 1, with southerly declinations and shallow up inclinations, is present in specimens from seven sample locations (Figures 2a and 2b and Table 1a). It has a mean intensity of 0.095 mA m^{-1} (standard deviation of 0.065; $n = 15$ sites). Component 2 is present in specimens from three locations, has northwesterly declinations and moderate down inclinations (Figures 2a and 2c and Table 1b), and has a mean intensity of 0.170 mA m^{-1} (standard deviation of 0.152; $n = 10$ sites). Possible antipodal directions for both components are found in specimens at some sites (Figures 2d and 2e).

[21] Component 1 is present at 16 of the sites sampled in the Confusion Range and Mountain Home Range (Table 1a). This component is commonly removed between 275° and 550°C (Figure 2b). Within a site, specimen directions are well grouped (Table 1a).

[22] Specimens from site CS3 have a component with northerly declinations and shallow down inclinations (Figures 2d and 3a) that decays between 380 and 420°C. A reversal test [*McFadden and McElhinny*, 1990] was performed on the specimen directions from sites CS1 and CS3. The deviation from antipodality is 15.8, which is less than 22.0, the 95% critical value, showing the two polarities to be insignificantly different. The results pass at classification Rc. The component in specimens from site CS3 could be an antipode to the reversed directions of component 1, although we note that only three specimens were found with the normal direction and more sampling would be needed to confirm this observation.

[23] Fold tests were conducted on the component 1 data from the Mountain Home Range and Confusion Range. An incremental fold test [*Watson and Enkin*, 1993] on the plunge-corrected site means (Table 1a) from the anticline-syncline couplet in the Confusion Range (Figure 3a) indicates that the best grouping is at 97 ± 15% unfolding (Figure 3b) with a declination of 172.8° and an inclination of −39.1° ($\alpha_{95} = 9.5°$, $k = 41.1$, Table 2). We note, however, that there is only one site from the easterly limb of the fold. A second incremental fold test was performed on the plunge-corrected site means (Table 1a) from the Mountain Home Range syncline (Figure 4a) and indicates a best grouping at 92 ± 9% unfolding (Figure 4b) with a declination of 171° and an inclination of −17.2° ($\alpha_{95} = 8.3°$, $k = 46.1$, Table 2). Because a few sites used in the fold tests had low specimens numbers or high α_{95} values, both fold tests were analyzed after removing all sites with less than four specimens and/or with α_{95} values greater than 20°. The results were similar to the unfiltered fold test results (Table 2). Block rotation Fisher analysis [*Enkin and Watson*, 1996] fold tests were performed and also indicate a prefolding magnetization (Table 2). The results from the fold tests include 100% unfolding within the error limits. Therefore the fold tests suggest that component 1 is prefolding. Plotting the pole positions for the stratigraphic directions (100% unfolding) on the APWP for North America [*Van der Voo*, 1993] indicates that component 1 is Late Triassic to Early Jurassic in age (Figure 5).

[24] The Tertiary tilt observed in the Confusion Range and Mountain Home Range [*Nichols et al.*, 2002] was

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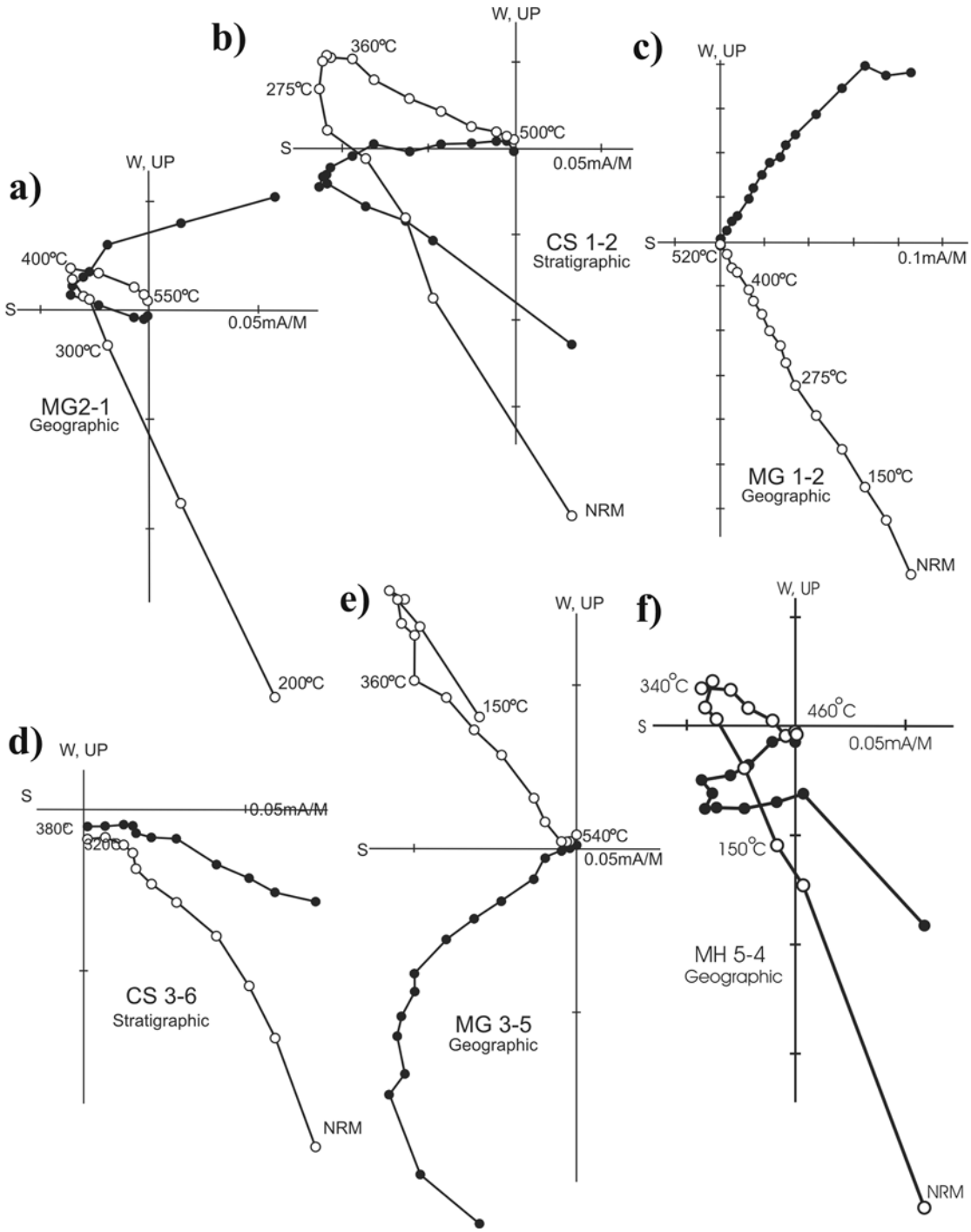


Figure 2. Representative orthogonal projections of thermal demagnetization data from different locations in Utah showing the remanent components: (a) VRM (200–300°C), component 1 (300–400°C) and component 2 (400–550°C), with NRM removed because of scale, (b) component 1 (360–500°C), (c) component 2 (275–520°C), (d) possible antipode to component 1 (320–380°C), (e) possible antipode to component 2 (360–540°C), with NRM removed because of scale, and (f) VRM (150–340°C) and component 1 (340–460°C). The solid circles represent the horizontal component, and the open circles represent the vertical component.

accounted for by correcting the geographic directions for the maximum tilt, correcting for the plunge, and then performing the fold test. The tilt correction does not significantly change the grouping of the site means. The tilt corrected pole positions were compared with the uncorrected poles on the APWP. Removing the maximum of 25° of westerly tilt in the Confusion Range gives a new pole position of 58.7°N, 118.4°E that is close to the uncorrected pole. Removing 25° of easterly tilt in the Mountain Home Range gives a pole position of 58.8°N, 59.2°E that does not

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Table 1a. Site Means for Specimens With Component 1 Directions[a]

Location/Site	Strike	Dip	N/N_o	Geographic				Stratigraphic			
				D, deg	I, deg	k	α_{95}, deg	D, deg	I, deg	k	α_{95}, deg
Confusion Range (39.2°N, 113.7°W)											
CS 1	13	12E	7/7	173.9	−14.8	183.0	4.5	177.4	−18.4	181.0	4.5
CS 3[b]	37	9E	3/6 (3)	10.1	13	27.3	24.1	12.4	24.8	12.4	36.5
BW 1	186	55W	6/6	214.3	−29.3	166.2	5.2	173.8	−38.3	168.1	5.2
BW 2	179	57W	3/3	203.0	−27.8	54.9	16.8	165.4	−33.8	55.0	16.8
BW 3	177	56W	5/5	209.5	−25.1	110.0	7.3	171.1	−39.8	109.7	7.3
BW 4	177	56W	4/4	219.2	−27.9	769.5	3.3	172.2	−48.9	766.6	3.3
SS 2	180	38W	7/7	193.1	−43.9	344.0	3.3	157.0	−40.3	344.3	3.3
SS 3	180	38W	5/6	219.8	−34.7	83.0	7.4	185.8	−50.6	68.3	9.3
Mountain Home Range (38.6°N, 113.9°W)											
MH 4	321	69E	6/6	138.0	−19.6	133.3	5.8	159.8	−9.6	133.4	5.8
MH 5	321	69E	5/5	143.6	−26.3	122.5	6.9	167.9	−6.9	122.7	6.9
MH 6	321	69E	5/5	140.1	−20.5	189.8	5.6	161.3	−8.0	193.6	5.5
MG 2	177	30W	5/5	194.3	−19.2	98.3	7.8	184.4	−25.2	97.9	7.8
MG 5	175	29W	4/4	184.1	−24.1	510.0	4.1	172.7	−25.3	521.0	4
MG 7	178	29W	3/5	186.4	−8.7	91.6	13	183.5	−11.7	91.8	12.9
MG 9	198	18W	4/4	179.0	−25.6	97.9	9.3	174.4	−18.7	97.7	9.3
MG 10	180	16W	4/4	170.7	−27.4	218.0	6.2	164.6	−23.8	215.6	6.3
Stansbury Mountains (40.6°N, 112.6°W)											
SM2	202	50W	2/5	176.5	−54.9	-	-	151.2	−19.7	-	-
SM4	333	50E	4/5	137.8	13.4	21.5	20.3	137.5	−2.6	21.4	20.3
Johnson Pass (40.2°N, 112.4°W)											
JNP 1	85	45E	8/9	142.6	−8.1	79.8	6.2	127.8	−43.7	79.8	6.2
JNP 2	85	45E	8/8	134.3	−8.5	23.8	11.6	118.4	−39.3	23.8	11.6
JNP 3	85	45E	7/7	131.3	0.2	42.1	9.4	121.6	−30.6	42.2	9.4

[a]The stratigraphic data from the Confusion Range, Mountain Home Range, and Stansbury Mountains have been plunge corrected. Strike and dip are the strike and dip of the bedding, N/N_o is the ratio of the number of specimens analyzed to the total number of specimens and the number in parentheses is the number of specimens with normal directions, k is the precision parameter related to the deviation of the true mean of the magnetic direction in the specimens [*Fisher*, 1953], α_{95} is the radius of 95% probability around the calculated true mean.
[b]Possible component 1 antipode.

significantly change the inferred age of the component. Since the declinations of component 1 are similar to the trend of the faults, removing the tilt does not significantly change the inferred age of the magnetizations (Figure 5).

[25] Analysis of preliminary sample sets collected in the Stansbury Mountains and Johnson Pass indicates that they contain specimen directions similar to component 1 (Table 1a). A fold test, based on the specimen directions

Table 1b. Site Means for Specimens With Component 2 Directions[a,b]

Location/Site	Strike	Dip	N/N_o	Geographic				Stratigraphic			
				D, deg	I, deg	k	α_{95}, deg	D, deg	I, deg	k	α_{95}, deg
Mountain Home Range (38.6°N, 113.9°W)											
MG1	167	34W	5/5	305.8	45.1	470.0	3.5	293.1	19.1	469.0	3.5
MG2	177	30W	3/5	314.8	57.8	108.2	11.9	297.8	33.6	107.4	12.0
MG3[b]	174	32W	4/5 (4)	124.8	−38.3	39.9	14.7	118.0	−12.2	39.9	14.7
MG4A	170	37W	4/4	8.6	47.7	59.1	12	329.2	46.0	59.0	12.1
MG5	175	29W	2/4	344.6	61.4	-	-	310.4	46.6	-	-
MG6	175	30W	4/4	248.7	68.5	56.1	12.4	259.7	39.0	55.8	12.4
MG7	178	29W	3/5	286.2	67.5	23.1	26.3	279.3	39.2	23.0	26.3
MG9	198	18W	2/4	319.6	41.6	-	-	316.0	25.8	-	-
Pavant Range (38.6°N, 112.6°W)											
CF1	8	33E	2/3	353.4	67.1	-	-	56.1	55.7	-	-
CF2	8	33E	2/3	58.7	71.0	(179.0)	(18.7)	82.2	41.0	(178.0)	(18.8)
CF3	8	33E	5/5	14.0	68.2	208.0	5.3	63.5	49.2	209.0	5.3
Oquirrh Mountains (40.3°N, 112.2°W)											
OPR1	145	40W	9/10 (10)	126.5	−42.1	13.6	14.5	104.0	−21.2	13.6	14.5
OPR2	145	40W	5/7 (6)	116.7	−32.5	29.9	14.2	103.7	−8.9	29.7	14.3

[a]The stratigraphic data from the Mountain Home Range have been plunge corrected. Strike and dip are the strike and dip of the bedding, N/N_o is the ratio of the number of specimens analyzed to the total number of specimens and the number in parentheses is the number of specimens with reversed directions, k is the precision parameter related to the deviation of the true mean of the magnetic direction in the specimens [*Fisher*, 1953], α_{95} is the radius of 95% probability around the calculated true mean.
[b]Possible component 2 antipode.

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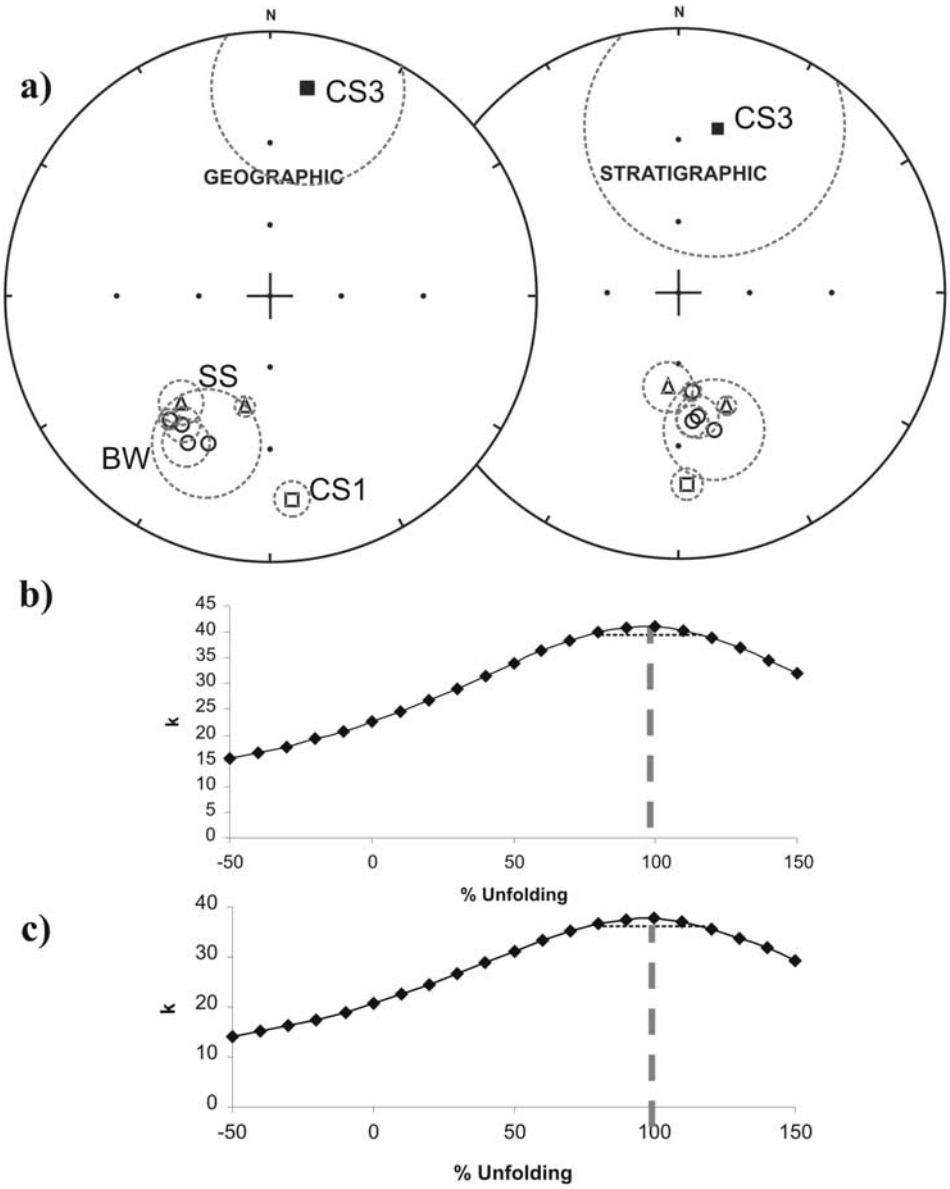


Figure 3. (a) Equal-area projection for component 1 site means in the Confusion Range in both geographic and stratigraphic positions (with α_{95} circles for site means). Site CS (squares) is on the east dipping limb, sites SS (triangles) and BW (circles) are on the west dipping limb. The CS3 site mean was not included in calculation of the α_{95}. The open symbols represent negative inclinations. (b) Graph showing the percent unfolding versus grouping (k) for component 1 in the Confusion Range. Although this test produces multiple synthetic curves, we only present the best grouping result. The best grouping is at 97% unfolding ±15% [*Watson and Enkin*, 1993]. (c) Graph showing the percent unfolding versus grouping (k) for the block rotation Fisher analysis test [*Enkin and Watson*, 1996]. The vertical dashed lines highlight the best grouping unfolding position and the horizontal dashed lines for Figures 3b and 3c show the error bars for the fold tests.

from two sites in the Stansbury Mountains, suggests a prefolding magnetization (Table 2). The specimen directions from the Johnson Pass location, which is located in a fault zone at the southern end of the Stansbury Mountains, have more easterly declinations than component 1. This could be due to rotations caused by the extensive faulting in the area. A regional block rotation Fisher analysis fold test on the Stansbury Mountains, Johnson Pass, Mountain Home Range, and Confusion Range sites suggests a prefolding time of remanence acquisition (Table 2).

[26] Component 2, with northerly and northwesterly declinations and moderate down inclinations is present in 10 of the sites in the Mountain Home Range and Pavant Range (Table 1b). In specimens that contain only component 2, it is removed between ~300 and 520°C (Figure 2c). However, in specimens that contain both components, unblocking of component 2 occurs between 360 and 460°C (Figure 2a). One site (MG3, Figures 2e and 6 and Table 1b) contains a component with southeasterly declinations and moderate up inclinations.

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Table 2. Summary of Paleomagnetic Data for Components 1 and 2 With the Results From the Three Fold Tests and One Regional Fold Test[a]

Location	N/N_o	Geographic				Stratigraphic				Pole	β_{95}
		D, deg	I, deg	k	α_{95}, deg	D, deg	I, deg	k	α_{95}, deg		
Component 1											
Mountain Home Range	8/8	167.3	−22.9	13.5	15.6	**170.9**	**−16.3**	**45.4**	**8.3**	**58.6°N, 83.6°E**	**6.1**
Best grouping (92 ± 9% unfolding)						171.0	−17.2	46.1	8.3	59.1°N, 83.6°E	6.2
Mountain Home Range (sites, $n > 3$)	7/8	164.2	−24.8	14.1	16.6	**167.3**	**−16.9**	**50.7**	**8.6**	**57.9°N, 90.3°E**	**6.4**
Best grouping (89 ± 10% unfolding)						167.4	−18.3	57.5	8.4	58.7°N, 90.6°E	6.3
Block Rotation Fisher Test	8/8	167.3	−22.8	12.6	10.9/−11.4	**169.1**	**−16.3**	**44.7**	**5.6/−5.9**	**58.2°N, 86.9°E**	**4.1**
Best grouping (94 ± 9% unfolding)							−17.3	44.8	5.6/−5.9		
Tertiary tilt correction (25°)	8/8	178.8	−26.0	13.5	15.6	**183.6**	**−14.8**	**43.61**	**8.5**	**58.8°N, 59.2°E**	**6.2**
Confusion Range	7/8	204.6	−30.0	22.5	13.0	**171.6**	**−38.9**	**41.1**	**9.5**	**71.3°N, 91.3°E**	**8.7**
Best grouping (97 ± 15% unfolding)						172.8	−39.1	41.1	9.5	71.8°N, 88.2°E	8.8
Confusion Range (sites, $n > 3$)	6/8	204.9	−30.4	18.9	15.8	**172.7**	**−39.7**	**36.5**	**11.2**	**72.2°N, 88.9°E**	**10.4**
Best grouping (102 ± 14% unfolding)						171.9	−39.6	36.5	11.2	71.9°N, 91.1°E	10.4
Block rotation Fisher test	7/8	204.6	−29.9	20.7	8.9/−9.3	**171.6**	**−38.8**	**37.6**	**6.5/−6.8**	**71.3°N, 91.3°E**	**6.0**
Best grouping (96 ± 13% unfolding)							−38.8	37.6	6.5/−6.8		
Tertiary tilt correction (25°)	7/8	188.3	−37.3	22.5	13	**154.6**	**−31.5**	**39.8**	**9.7**	**58.7°N, 118.4°E**	**8.1**
Stansbury Mountains	6/10[b]	146.6	−8.9	4.0	38.1	**142.0**	**−8.5**	**21.3**	**14.8**	**40.2°N, 120.9°E**	**10.6**
Best grouping (121 ± 22% unfolding)						141.8	−7.6	25.1	13.7	39.7°N, 120.7°E	9.8
Regional (MHR, CR, JNP, SM)											
Block rotation Fisher test	19/21	173.1	−22.8	6.4	10.7/−10.9	**161.4**	**−28.5**	**12.2**	**7.5/−7.7**	**61.0°N, 105.7°E**	**6.1**
Best grouping (93 ± 7% unfolding)							−28.6	12.2	7.5/−7.7		
Component 2											
Mountain Home Range	7/8	318.6	60	13.9	16.8	297.7	37.5	15.2	16		
Pavant Range	3/3	19.4	70.8	48.9	17.8	68.5	49.2	48.9	17.8		
Regional fold test (sites, $a_{95} < 20$)	6/11	336.0	68.3	9.7	22.6	**316.2**	**60.3**	**2.8**	**49.7**	68.9°N, 202.1°E[c]	34.9
Block rotation Fisher test	6/11	336.0	67.0	8.7	22.9/−24.9	**316.2**	**56.5**	**2.5**	**26.3/−31.2**	69.7°N, 197.5°E	34.5
Best grouping (−44 ± 5% unfolding)			67.4	12.1	12.0/−11.5						
Tertiary tilt correction (25°)	6/11	19.3	69.7	14.4	18.3					70.1°N, 282.0°E	29.0

[a] N/N_o is the ratio of the number of sites analyzed to the total number of sites used in the statistics, k is the precision parameter related to the deviation of the true mean of the magnetic direction in the specimens [*Fisher*, 1953], α_{95} is the radius of 95% probability around the calculated true mean, β_{95} is the radius of the 95% confidence circle about the mean pole. Confidence interval from the *Watson and Enkin* [1993] fold test is after the best grouping percent. Block rotation Fisher analysis test from *Enkin and Watson* [1996]. Bold values are for 100% unfolding. Possible antipodes were not used in the fold tests.
[b] Specimen directions used instead of site means.
[c] Pole for the Geographic coordinates.

[27] Specimens from the Oquirrh Mountains also contain a component with southeasterly declinations and moderate up inclinations (Table 1b and Figure 6). Although similar in direction to MG3 from the Mountain Home Range, the Oquirrh Mountains have experienced higher burial temperatures than the other study areas and this component will be considered separately.

[28] A reversal test [*McFadden and McElhinny*, 1990] on the results from site MG3 and the other MG sites with α_{95} less than 20°. The deviation from antipodality is 24.2, which is less than 35.9, the 95% critical value, showing the two polarities to be insignificantly different. However, since the deviation is >20, the reversal test is assessed as indeterminate. More important than the statistical test, however, is the observation that there are two polarities in evidence, which has implications on the timing and duration of the remagnetization event.

[29] A regional fold test on component 2 sites with α_{95} values less than 20° from the Mountain Home Range and the Pavant Range indicates that the grouping is better in geographic than stratigraphic coordinates but the site directions are never grouped (Figure 6a and Table 2). A block rotation fold test indicates a best grouping at −44% unfolding (Figure 6b and Table 2). The reason for the −44% best grouping is unknown, but it is probably related to the incomplete separation of components and contamination of the direction in some specimens. Because of this, and the relatively low number of sites that contain this component, an age determination should be viewed with caution. Acknowledging this issue, we suggest that component 2 is a postfolding magnetization. Plotting the pole (Table 2) on the APWP for North America [*Van der Voo*, 1993] indicates that component 2 is Cretaceous to early Tertiary in age (Figure 5).

[30] Correcting component 2 for the maximum Tertiary tilt (25°) in the Mountain Home Range and then performing the regional fold test gives a declination of 19.3° and an inclination of 69.7° (α_{95} = 18.3°, k = 14.4). The grouping only improves slightly after correction for the Tertiary tilt (k = 14.4 versus k = 13.9). The new pole position (70.1°N, 282.0°E) is consistent with a Cretaceous to early Tertiary time of remanence acquisition (Figure 5).

[31] Sites CS2, CS4, SS1, SS4, SS5, MG4b, MG8, SM1, and SM3 are not listed in Table 1 because they have very noisy decay and/or only contain a VRM component. Sites MH1, MH2, and MH3 contain a remanent magnetization with directions similar to component 1 in the other MH sites. However, these sites are not listed in Table 1 because

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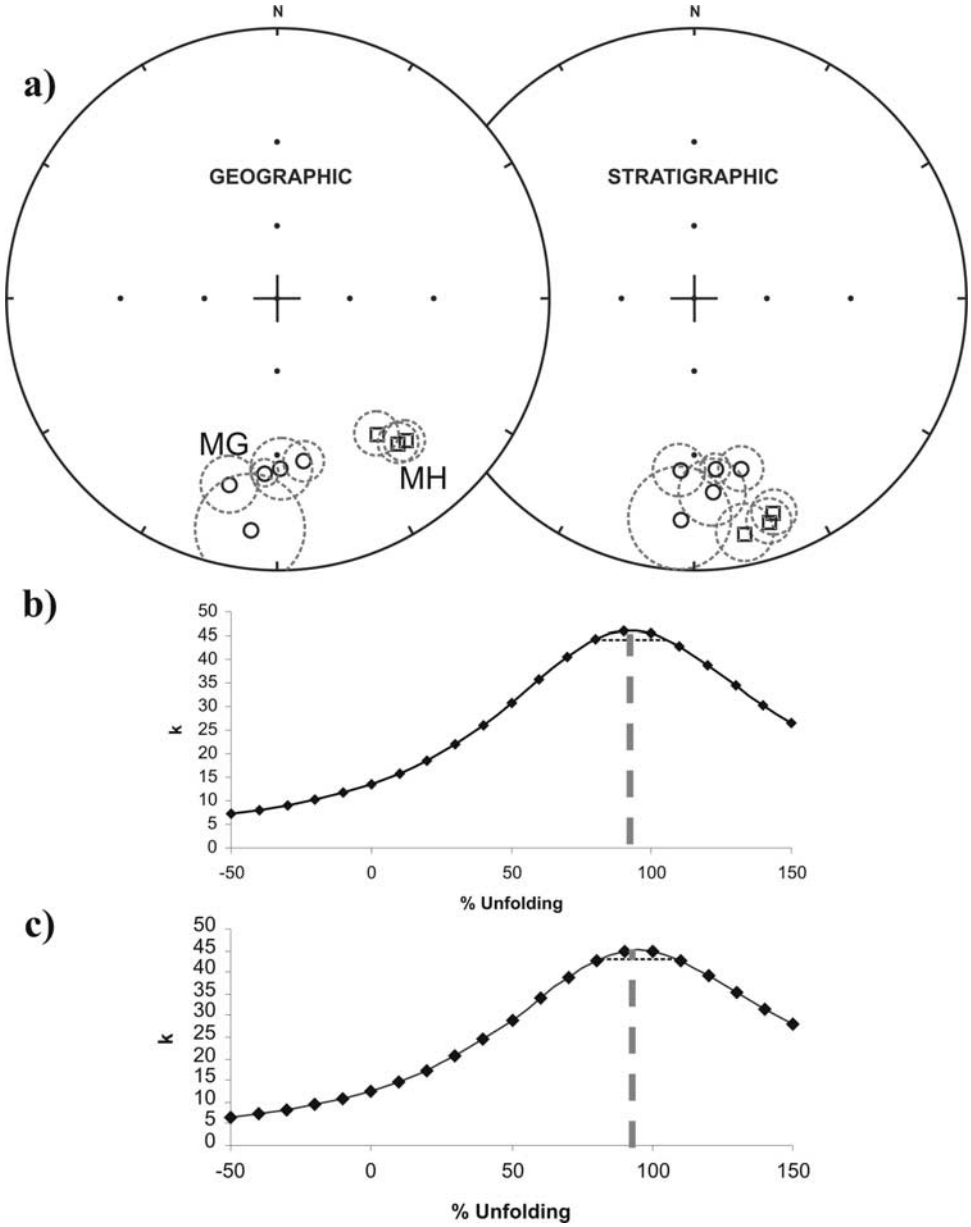


Figure 4. (a) Equal-area projection for component 1 site means in the Mountain Home Range (MG sites, circles; MH site, squares) in both geographic and stratigraphic positions (with α_{95} circles for site means). The open symbols represent negative inclinations. (b) Graph showing the percent unfolding versus grouping (k) for component 1 in the Mountain Home Range [*Watson and Enkin*, 1993]. Although this test produces multiple synthetic curves, we only present the best grouping result. The best grouping is at 92% unfolding ±9%. (c) Graph showing the percent unfolding versus grouping (k) for a block rotation Fisher analysis test. The vertical dashed lines highlight the best grouping unfolding position, and the horizontal dashed lines for Figures 4b and 4c show the error bars for the fold tests.

the specimen directions consistently have MAD angles greater than 15°.

4.2. Rock Magnetism

[32] Low-temperature analysis of 12 representative specimens shows a decrease in NRM intensity of 12.3 ± 6% after being subjected to liquid nitrogen. Thermal demagnetization of these specimens does not show distinct directions when compared to specimens that were not subjected to liquid nitrogen. The small decrease in intensity and the fact that the directions do not change suggest that multidomain grains with a modern VRM overprint are not an important contributor to the NRM [e.g., *Dunlop and Argyle*, 1991; *Halgedahl and Jarrard*, 1995].

[33] Thermal demagnetization of specimens suggests that the magnetization is contained in magnetite because decay is always complete at temperatures less that 580°C. This is also supported by the results of AF demagnetization, IRM acquisition, backfield measurements and triaxial thermal decay. In most specimens, AF demagnetization of the

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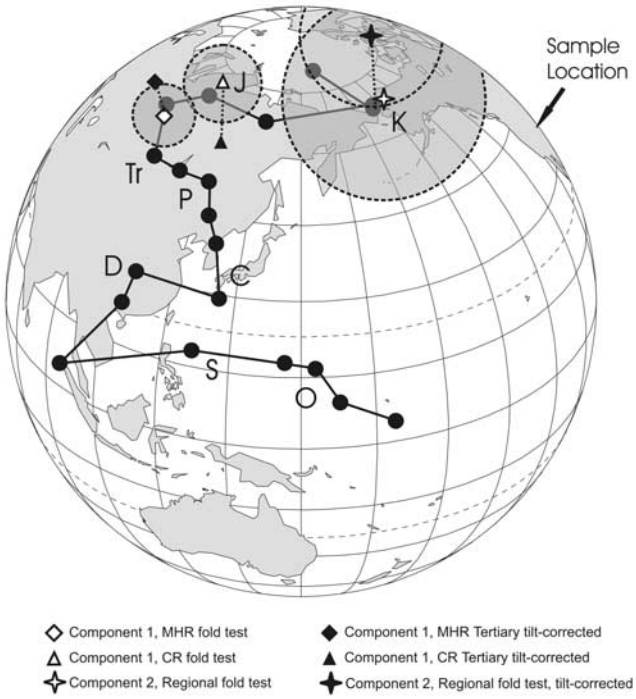


Figure 5. North American apparent polar wander path [*Van der Voo*, 1993] showing the poles for the fold test results. The results from the Tertiary tilt corrections are also shown. The error circles are the β_{95} values.

NRM up to 120 mT shows a significant amount of decay (Figure 7a), which suggests that the magnetization is contained in a low-coercivity phase. The IRM acquisition up to 2500 mT for most specimens shows saturation by 300 mT (Figure 7b) that is typical for magnetite. Backfield measurements also show an initial decrease in intensity and then resaturation by −300 mT (Figure 7b). The AF demagnetization up to 120 mT after IRM acquisition shows decay similar to that of the NRM (Figure 7a). Finally, the triaxial thermal decay of most specimens shows that the low-coercivity axis dominates and decays by 580°C (Figure 7c), indicating the presence of magnetite.

[34] Magnetite is the dominant magnetic phase in samples from all areas except the Mountain Home Range (MG and MH). The rock magnetic results indicate that ~50% of the specimens from this group contain evidence for the presence of hematite as well as other phases. The AF demagnetization of the NRM of these specimens shows significant decay by 120 mT (Figure 8a). The IRM acquisition curve for a representative specimen from this group, however, shows a rapid rise by 300 mT with a more gradual rise at higher fields (Figure 8b), suggesting the presence of both high- and low-coercivity phases. Backfield measurements for these samples show a decrease in intensity and do not resaturate by 1200 mT (Figure 8b). Triaxial thermal decay of these specimens shows that the 2500 mT axis dominates and decays at ~680°C, indicating the presence of hematite (Figure 8c). Compared to the AF demagnetization of NRM, the AF treatment after IRM acquisition shows much less decay (Figure 8a). Specimens that show this difference in AF demagnetization typically decay 84 ± 14% during AF demagnetization of the NRM and 28 ± 11% during AF demagnetization of the IRM. This suggests that a low-coercivity mineral such as magnetite is the dominant magnetic phase in the NRM. The other magnetic phases present are not seen until a field is applied to them. Therefore the directions obtained from thermal demagnetization probably reside in magnetite. The demagnetization data for specimens from site MH5 decay in the same range as specimens from other sites and do not indicate the presence of hematite (Figure 2f). A decrease in intensity on the 500 mT axis at ~320°C in the triaxial thermal decay curve indicates that some pyrrhotite could be present (Figure 8c). Limited decay between 275 and 320(350)°C in the orthogonal demagnetization plots (Figure 2) suggests that pyrrhotite could carry some of the component 1 magnetization. For a few specimens, triaxial thermal decay shows an abrupt decrease in intensity of the 2500 mT axis at 100°C, suggesting the presence of goethite.

[35] To further study the magnetic mineralogy, the IRM acquisition and the AF demagnetization of the IRM were compared in crossover plots [*Cisowski*, 1981]. For most specimens, the curves intersect between 40 and 50% normalized moment (Figure 7d), suggesting noninteracting single-domain (SD) and/or pseudosingle-domain (PSD) particles [*Cisowski*, 1981]. In addition, crossover typically occurs between 40 and 60 mT (Figure 7d). This behavior suggests that magnetite is the dominant remanence carrier [*Symons and Cioppa*, 2000]. The IRM decay curves of these specimens have inflected shapes (e.g., Figure 7a) which, according to *Dunlop* [1983], are representative of SD or PSD behavior. For specimens that contain higher coercivity minerals (i.e., goethite and hematite), as discussed above, the IRM acquisition and IRM decay curve do not show crossover.

4.3. Petrography

[36] Thin section analysis of specimens from western and central Utah shows that the sampled lithologies are mainly wackestones and mudstones. Most specimens have very thin (less than 100 μm wide) calcite-filled veins and contain small to moderate amounts of authigenic hematite and unaltered pyrite. In some specimens, it appears that pyrite grains were altered by an iron oxide, interpreted to be magnetite (Figure 9).

4.4. Geochemistry

[37] The $\delta^{18}O$ (PDB) and $\delta^{13}C$ (PDB) values for the 35 carbonate samples measured are similar, with only a few outliers. In the Confusion Range and Mountain Home Range the $\delta^{18}O$ values range from −6 to −4‰ and $\delta^{13}C$ values range from −1.5 to 3‰. The samples from the Pavant Range exhibit a greater range with the $\delta^{18}O$ values from −10 to −2‰ and have $\delta^{13}C$ values from about 1 to 4‰. However, all 35 values fall within the range for common marine limestones [e.g., *Hudson*, 1977] (Figure 10).

[38] Radiogenic isotope analysis of 12 carbonate samples yields $^{87}Sr/^{86}Sr$ values ranging from 0.707649 to 0.708112 (Figure 11). These values are consistent with seawater curves constructed from samples from the southern interior of the United States for Mississippian age limestones [*Denison et al.*, 1994]. Therefore these samples closely represent the $^{87}Sr/^{86}Sr$ values of seawater at the time the

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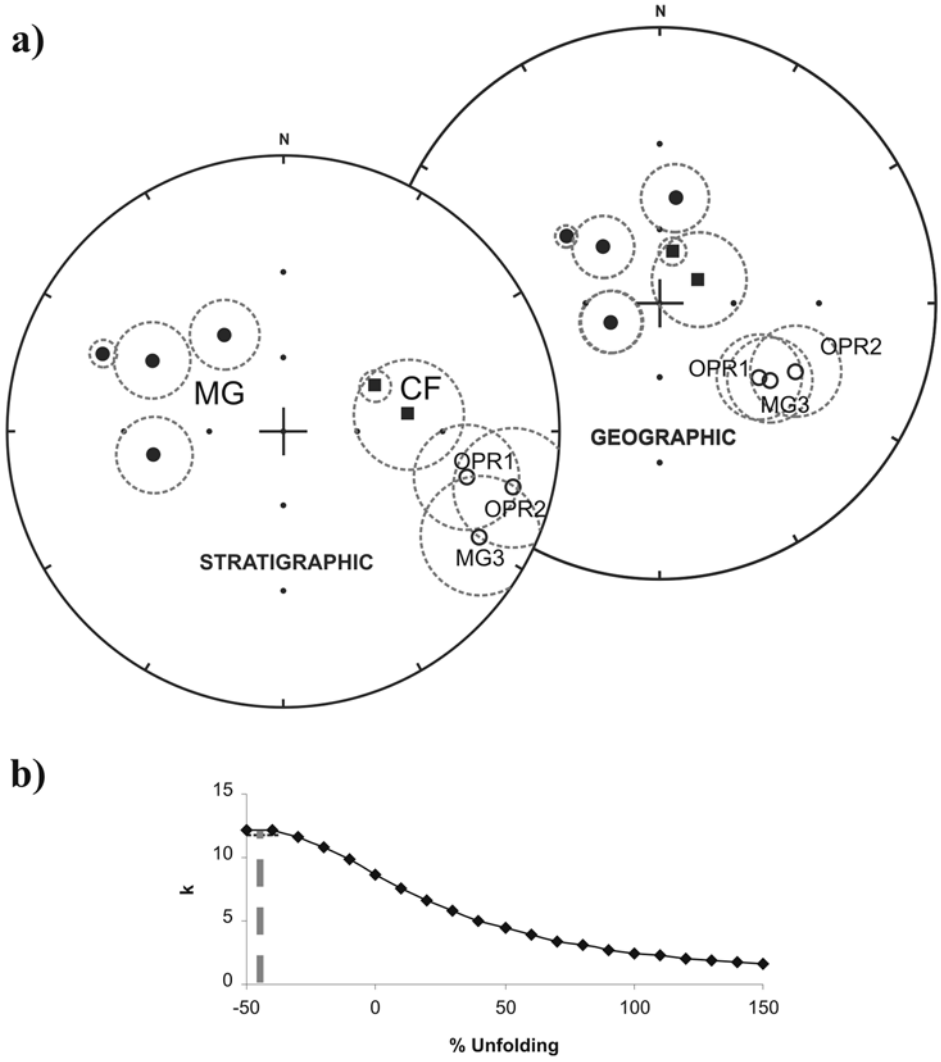


Figure 6. (a) Equal-area projections for component 2 site means (with α_{95} circles). Regional fold test comparing geographic and stratigraphic site means from MG (circles) and CF (squares) sites are shown. The site mean of possible antipode to component 2 (site MG3) and the OPR site means are shown but were not used in the regional fold test. (b) Graph showing the percent unfolding versus grouping (k) for the block rotation Fisher analysis regional fold test. The vertical dashed line highlights the best grouping unfolding position, and the error bar is shown by the horizontal bar. The open symbols represent negative inclinations, and solid symbols represent positive inclinations.

Deseret Limestone and Chainman Shale were deposited. This similarity suggests that the rocks were not extensively altered by externally derived fluids that are commonly enriched in ^{87}Sr [e.g., *Stueber et al.*, 1984; *Chaudhuri et al.*, 1987; *Clauer et al.*, 1989].

4.5. Mathematical Modeling for the Formation of Illite in the Deseret Limestone

[39] The model calculation shows that the percentage of illite in I-S increases from 5% to 100% between 287 and 216 Ma. According to burial curves [*Huntoon et al.*, 1999], the period of 287 to 216 Ma corresponds to burial up to 2.5–3.0 km and burial temperatures no greater than 75°C. The percentage of illite in I-S remains at 100 percent from ~216 Ma to present even though the Deseret Limestone is buried to 12 km. The calculated mean age of illitization is 247 Ma, slightly younger than the midpoint of the period between 216 and 287 Ma.

5. Discussion

[40] The samples of the Mississippian Deseret Limestone and Chainman Shale collected in western and central Utah contain two ancient components. On the basis of fold tests, one component is a prefolding magnetization that was acquired in the Late Triassic to Early Jurassic based on the pole position. The other component, based on a regional fold test, is apparently a postfolding magnetization that is Cretaceous to early Tertiary in age. The limestones at all sampling locations that were used for the fold tests have CAI values of 1.5 to 2 and have not been heated to temperatures above 140°C [*Sandberg et al.*, 1980]. These

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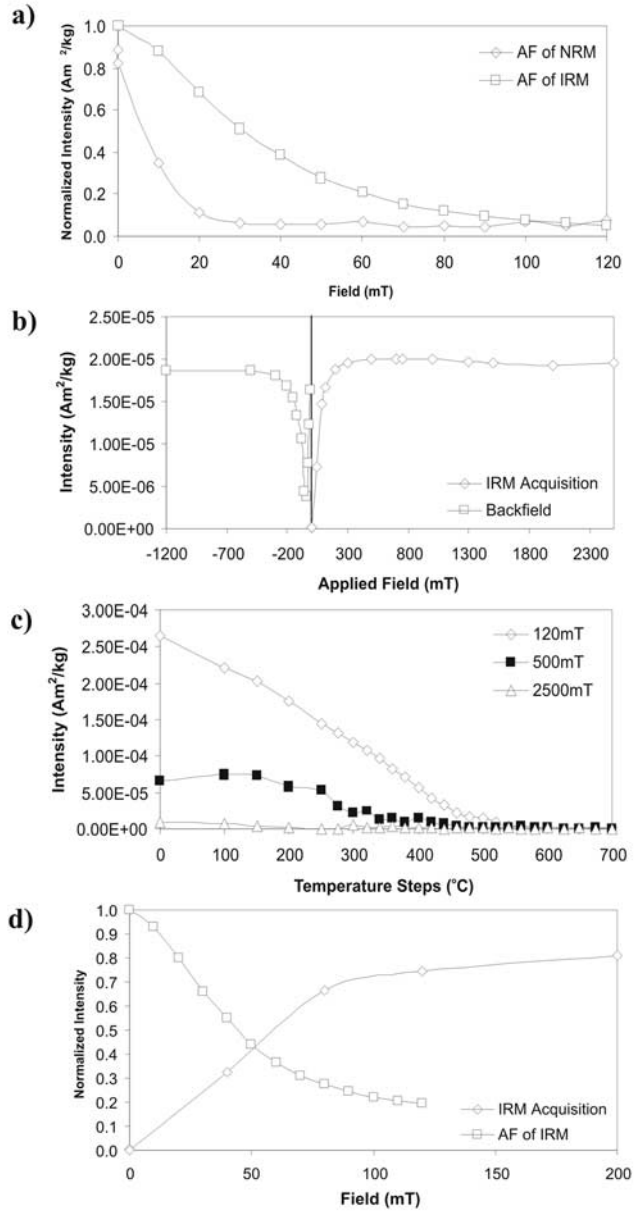


Figure 7. Specimen CS3-1b (a) AF demagnetization of NRM and IRM, (b) IRM acquisition and backfield, (c) triaxial thermal decay of IRM, and (d) specimen BW1-7 IRM acquisition and AF demagnetization of the IRM [*Cisowski*, 1981].

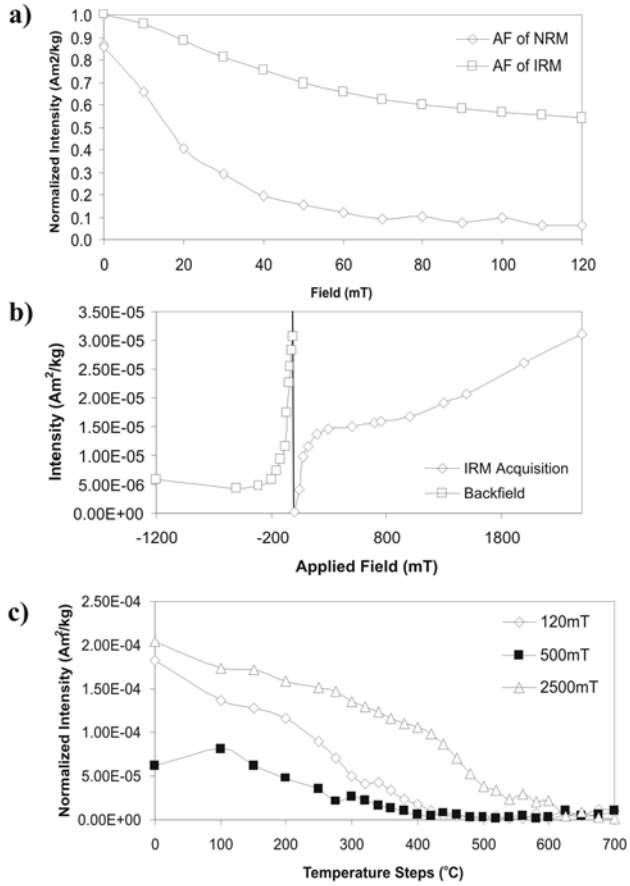


Figure 8. Specimen MH5-6a (a) AF demagnetization of NRM and IRM, (b) IRM acquisition and backfield, and (c) triaxial thermal decay of IRM.

burial temperatures are too low for these remagnetizations to be thermal viscous remanent magnetizations (TVRMs) [*Pullaiah et al.*, 1975]. Therefore the remagnetizations are interpreted as CRMs.

[41] The Oquirrh Mountains have been heated to temperatures greater than 300°C, which is significantly higher than the temperatures at the other locations used in the fold tests. The specimens from the Oquirrh Mountains typically have maximum unblocking temperatures between 500 and 580°C. These high unblocking temperatures suggest that the component in these sites could be a CRM [e.g., *Pullaiah et al.*, 1975] although this interpretation should be treated with caution because the maximum burial temperature is not known. Localized heating to high temperatures is possible in the Oquirrh Mountains; *Melker and Geissman* [1997] report a primary magnetization in the Tertiary intrusive and extrusive igneous rocks in the Oquirrh Mountains. Because

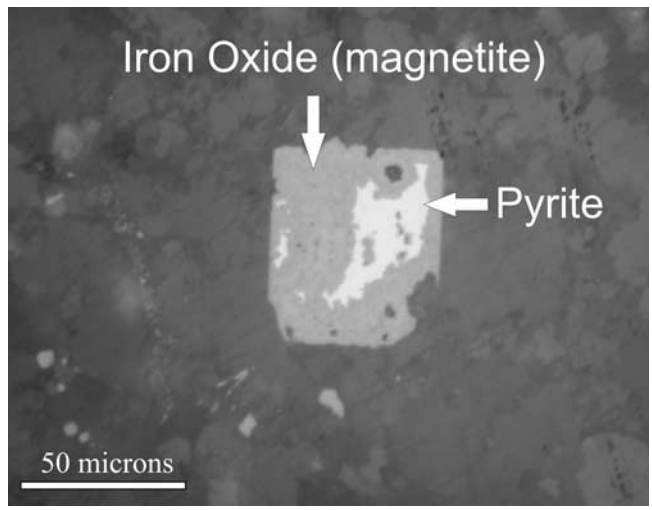


Figure 9. Thin section showing an iron oxide (magnetite?) replacing pyrite in reflected light for sample SS2-3.

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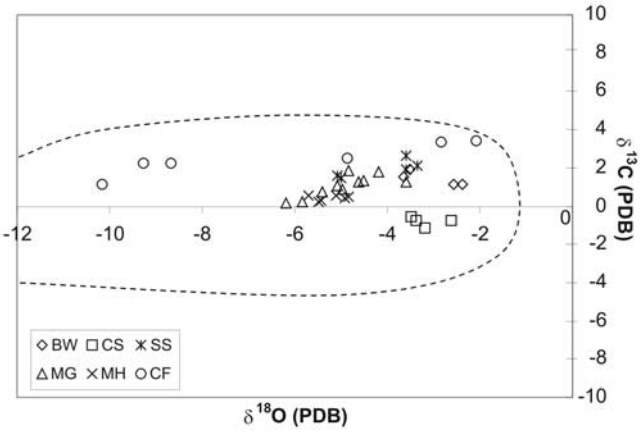


Figure 10. Plot of stable isotope data for the 35 carbonate samples. The different symbols correspond to the locations from which the samples were collected. The dashed line represents the range of common marine limestones after *Hudson* [1977].

of the uncertainty about the origin of this magnetization, it is not grouped with component 2.

[42] Several sites contain specimen directions that could be antipodal to components 1 or 2, although the data are limited and more sampling would be needed for confirmation. If antipodal components are present, then the CRMs were acquired over long enough periods to experience both normal and reversed polarities.

[43] The results from the rock magnetic analysis indicate that both CRMs reside in SD and/or PSD magnetite. Small amounts of pyrrhotite, goethite, and hematite are present in some specimens although they do not appear to contribute to the remanent magnetization.

[44] Radiogenic isotope analysis shows that $^{87}Sr/^{86}Sr$ values for the samples are similar to $^{87}Sr/^{86}Sr$ values from seawater curves for Mississippian age carbonates (Figure 11). These similarities suggest that externally derived fluids did not extensively alter the rocks.

[45] The stable carbon and oxygen isotope values for the samples plot within the field for common marine limestones (Figure 10), with most close to the average reported by

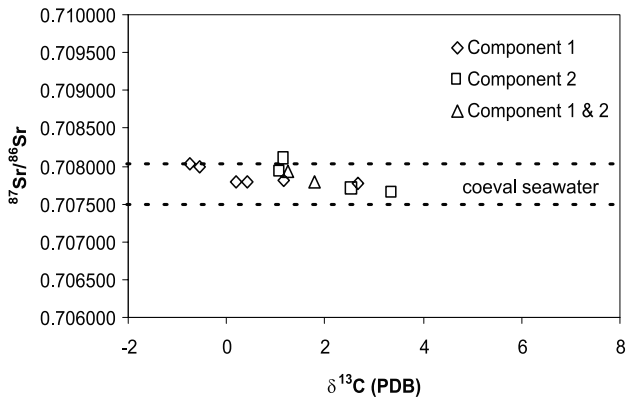


Figure 11. The $\delta^{13}C$ (PDB) versus $^{87}Sr/^{86}Sr$ values for 12 samples. The range of $^{87}Sr/^{86}Sr$ values for Mississippian coeval seawater is shown [*Denison et al.*, 1994].

Keith and Weber [1964] for average Mississippian age marine limestones (i.e., $\delta^{18}O \sim -5‰$ and $\delta 13C \sim 2‰$). The dissolution and recrystallization of marine limestones by externally derived fluids can result in ^{13}C depletion [e.g., *Gao et al.*, 1992]. This depletion is not observed in the present study, consistent with the interpretation that the rocks with the CRMs are unlikely to have experienced significant alteration by externally derived fluids.

[46] *Jewell et al.* [2000] reported that rocks deposited during the Delle phosphatic event are depleted in ^{18}O by 3–5‰ relative to the overlying and underlying open marine limestones. They interpret this isotopic excursion as being indicative of a change in depositional environment (shallow water) in which ^{18}O depletion reflects either brackish or meteoric water. In the present study, samples from the Pavant Range (CF, Figure 10) exhibit this ^{18}O depletion observed by *Jewell et al.* [2000]. However, we did not observe a similar depletion in ^{13}C, which they attributed to a high organic carbon input [*Jewell et al.*, 2000].

[47] Petrographic analyses identified pyrite grains being altered to an iron oxide interpreted to be magnetite. Previous studies have also noted pyrite grains altered to an iron oxide in carbonate rocks that contain CRMs [e.g., *Fruit et al.*, 1995; *Weil and Van der Voo*, 2002]. Although the authigenic magnetite observed in thin sections in this study (Figure 9) is probably too coarse to carry the remanence, similar finer grained magnetite is inferred to carry the CRMs.

[48] Comparing the ages of the CRMs to the time versus vitrinite reflectance curve from *Huntoon et al.* [1999] illustrates the relationship between the magnetizations and timing of the oil window (Figure 12). Component 1 (CRM1) plots at the beginning of the oil window. The age of CRM1 is prior to the onset of the Sevier orogeny and the geochemical analysis provides no evidence for fluid

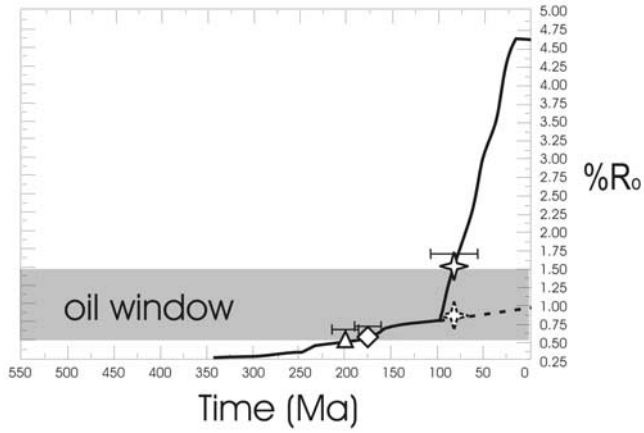


Figure 12. Time versus vitrinite reflectance curve for the Mississippian Deseret Limestone [modified from *Huntoon et al.*, 1999] (AAPG copyright 1999, with permission of the AAPG whose permission is required for further use) with the oil window. The triangle (CR) and diamond (MHR) represent the Late Triassic to Early Jurassic component 1, and the star represents the Cretaceous to early Tertiary component 2. Error bars for the poles were estimated from the APWP. The dashed line represents the estimated curve for the Chainman Shale in western Utah, and the dashed star represents component 2 on the estimated curve.

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alteration, suggesting that this CRM could be related to a burial diagenetic process. Component 2 (CRM2) plots toward the end of the oil window (Figure 12). This CRM is postfolding and the timing (Cretaceous to early Tertiary) coincides with the Sevier and Laramide orogenies. However, the geochemical ($^{87}Sr/^{86}Sr$ and $\delta^{18}O/\delta^{13}C$ values) suggest that externally derived fluids have not extensively altered the rock. This suggests that a burial diagenetic process is the origin for CRM2. It should also be noted that *Huntoon et al.* [1999] suggest that the time versus vitrinite reflectance curve is similar for western Utah and the Pavant range prior to 97 Ma. After the thrusting event in the Pavant Range at 97 Ma they state that the curve continues to increase but not as dramatically. Therefore comparison with the estimated curve for western Utah indicates that CRM2 plots within the center of the oil window rather than at the maximum limit (Figure 12). Since both CRMs plot within the oil window and show minimal if any geochemical or petrographic evidence for fluid alteration by external fluids, they are interpreted to result from burial diagenetic processes.

[49] Previous work has shown that organic matter at low to moderate burial temperatures may cause magnetite authigenesis and an associated CRM. For example, *Plaster-Kirk et al.* [1995] found a Carboniferous CRM residing in magnetite in organic-rich lacustrine deposits of the Middle Old Red Sandstone in Scotland that correlates with the suggested time of organic matter maturation. *Banerjee et al.* [1997] found a CRM in authigenic magnetite from carbonates of the Pennsylvanian Belden Formation in Colorado. The age of the CRM varies from late Paleozoic to Cretaceous and coincides with the modeled time of organic matter maturation in different parts of the basin [*Banerjee et al.*, 1997]. *Cioppa et al.* [2002] related a CRM in the ''Nordegg Member'' of the Fernie Formation in Canada to organic matter maturation of the organic-rich shales. In addition, *Brothers et al.* [1996] conducted simulation experiments to demonstrate that authigenic magnetite can form from pyrite in the presence of organically complex ferric iron that can act as an oxidizing agent at temperatures of $\sim$90°C.

[50] The smectite-to-illite transition occurs in the same temperature range as hydrocarbon generation [*Pevear*, 1999] and CRMs have been associated with the smectite-to-illite transition in which illitization releases iron to form authigenic magnetite [*Katz et al.*, 2000]. In Jurassic and Cretaceous carbonates in the Vocontian trough (southeast France), *Katz et al.* [2000] found a CRM carried by magnetite at locations where smectite had been altered to other clay minerals but the CRM was absent or weak in locations where a significant amount of smectite was still present. There was no geochemical evidence for alteration by orogenic fluids and burial diagenesis of smectite was interpreted to be the remagnetization mechanism [*Katz et al.*, 2000]. *Gill et al.* [2002] used a similar presence-absence test in the Overthrust Belt of Montana to show the connection between the timing of a CRM and the smectite-to-illite conversion. Laboratory simulation experiments are also consistent with a connection between clay diagenesis and magnetite authigenesis [*Hirt et al.*, 1993; *Cogoini et al.*, 2002].

[51] The modeled date for the smectite-to-illite transformation in the Deseret Limestone (median age 247 Ma) is $\sim$50 Myr prior to the age of CRM1, suggesting that clay diagenesis is not the remagnetization process. The range for illitization (287–216 Ma), however, overlaps with the age of CRM1. Because of this overlap, and the inherent uncertainty in the modeling, clay diagenesis cannot be ruled out as the remagnetization mechanism for CRM1. There is no overlap between the age of the CRM2 and the modeled age of illitization, and clay diagenesis is not considered a likely remagnetization process for that component.

[52] The origin of pervasive CRMs has been controversial and is unresolved. In addition to burial diagenetic processes, fluids activated as a result of orogenic activity [e.g., *Oliver*, 1992] are commonly hypothesized as an agent of remagnetization. In some cases the orogenic fluid model is a reasonable interpretation, particularly for those localized CRMs near or in conduits for flow [*Elmore et al.*, 1999, 2001; *Elmore*, 2001]. However, many units such as the Deseret Limestone contain pervasive CRMs but do not contain geochemical evidence for alteration by externally derived fluids [e.g., *Elmore et al.*, 1993, 2001]. These CRMs therefore are probably related to another widespread mechanism such as a burial diagenetic process. The agreement of the timing of remagnetization and maturation of organic matter in the Deseret Limestone and Chainman Shale suggest that the CRMs are probably associated with the maturation of organic matter or perhaps clay diagenesis for CRM1.

[53] The Triassic to Jurassic age for CRM1 is unusual compared to previous studies in North America where most CRMs are Late Paleozoic or Cretaceous in age [e.g., *McCabe and Elmore*, 1989; *Gillett and Van Alstine*, 1982; *Gillett and Geissman*, 1993]. Most of the magnetizations are single polarity, although several studies report dual polarity Permian to Triassic magnetizations in Early Paleozoic carbonates that reside in magnetite [e.g., *Gillett and Taylor*, 1986; *Stamatakos et al.*, 1998]. These magnetizations postdate the Late Paleozoic Superchron and a fold test on a Triassic component [*Gillett and Taylor*, 1986] indicates it is prefolding, similar to CRM1 reported in this study.

6. Conclusions

[54] There are two ancient magnetizations in the Mississippian Deseret Limestone and Chainman Shale in western and central Utah. Both components are secondary CRMs that reside in magnetite. Fold test results suggest that CRM1 is a prefolding magnetization that is Late Triassic to Early Jurassic in age and that CRM2 is probably a postfolding magnetization that is Cretaceous to early Tertiary in age. Comparing the ages of the CRMs to burial data, it appears that the prefolding and postfolding CRMs could be the result of burial diagenetic processes such as maturation of organic matter. Modeling of the smectite-to-illite transformation in the Deseret Limestone suggests an age that overlaps with the CRM1 and is prior to acquisition of CRM2. Clay diagenesis therefore is not considered a likely remagnetization process for CRM2 but it could be for CRM1. Geochemical and petrographic analyses do not indicate alteration by externally derived fluids.

[55] The results of this study support the hypothesis that pervasive CRMs are associated with widespread burial diagenetic mechanisms such as maturation of organic matter. In addition, the results suggest that paleomagnetism can

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be used to determine the timing of these burial diagenetic processes which, when combined with conventional maturation data (i.e., conodont alteration, vitrinite reflectance, and burial curves), would benefit exploration efforts and help determine the viability of a potential source rock.

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References

Aronson, J. L., and J. Hower (1976), The mechanisms of burial metamorphism of argillaceous sediment: 2. radiogenic argon evidence, *Geol. Soc. Am. Bull.*, *87*, 738–744.

Baer, J. L., R. L. Davis, and S. E. George (1982), Structure and stratigraphy of the Pavant Range, central Utah, in *Overthrust Belt of Utah: Symposium and Field Conference, Publ. 10*, pp. 31–48, Utah Geol. Assoc., Salt Lake City.

Banerjee, S., M. Engel, and R. D. Elmore (1997), Chemical remagnetization and burial diagenesis: Testing the hypothesis in the Pennsylvanian Belden Formation, Colorado, *J. Geophys. Res.*, *102*, 24,825–24,842.

Bird, P. (1998), Kinematic history of the Laramide orogeny in latitudes 35°–49°N, western United States, *Tectonics*, *17*, 780–801.

Brothers, L. A., M. H. Engel, and R. D. Elmore (1996), The late diagenetic conversion of pyrite to magnetite by organically complexed ferric iron, *Chem. Geol.*, *130*, 1–14.

Chaudhuri, S., V. Broedel, and N. Clauer (1987), Strontium isotopic evolution of oil-field waters from carbonate reservoir rocks in Bindley field, central Kansas, USA, *Geochim. Cosmochim. Acta*, *51*, 45–53.

Cioppa, M. T., D. T. A. Symons, and M. Flore (2002), Initial analysis of the Jurassic "Nordegg Member" Fernie Formation: Using paleomagnetism to date source rock thermal maturation, *Phys. Chem. Earth.*, *27*, 1161–1168.

Cisowski, S. (1981), Interacting vs. non-interacting single domain behavior in natural and synthetic samples, *Phys. Earth Planet. Inter.*, *26*, 56–62.

Clauer, N., S. Chaudhuri, and R. Subramanium (1989), Strontium isotopes as indicators of diagenetic recrystallization scales within carbonate rocks, *Chem. Geol.*, *80*, 27–34.

Cogoini, M., R. D. Elmore, and M. Engel (2002), Thermal treatment of clays resulting in magnetic mineral formation in a smectite: An analogy to low burial conditions, *Eos Trans. AGU*, *83*(47), *Fall Meet. Suppl.*, Abstract GP11A-1078.

Denison, R. E., R. B. Koepnick, W. H. Burke, E. A. Hetherington, and A. Fletcher (1994), Construction of the Mississippian, Pennsylvanian and Permian seawater $^{87}Sr/^{86}Sr$ curve, *Chem. Geol.*, *112*, 145–167.

Dunham, R. J. (1962), Classification of carbonate rocks according to depositional texture, *Mem. Am. Assoc. Pet. Geol.*, *1*, 108–121.

Dunlop, D. J. (1983), Determination of domain structure in igneous rocks by alternating field methods, *Earth Planet. Sci. Lett.*, *63*, 353–357.

Dunlop, D. J., and K. S. Argyle (1991), Separating multidomain and single-domain-like remanences in pseudo-single-domain magnetites (215–540 nm) by low-temperature demagnetization, *J. Geophys. Res.*, *96*, 2007–2017.

Elliott, W. C., and G. Matisoff (1996), Evaluation of kinetic models for the smectite to illite transformation, *Clays Clay Miner.*, *44*, 77–87.

Elmore, R. D. (2001), A review of palaeomagnetic dating on the timing and origin of multiple fluid-flow events in the Arbuckle Mountains, southern Oklahoma, *Pet. Geosci.*, *7*, 223–229.

Elmore, R. D., D. London, D. Bagley, and G. Gao (1993), Evidence for paleomagnetic dating of burial diagenesis by basinal fluids, Ordovician carbonates, Arbuckle Mountains, southern Oklahoma, in *Applications of Paleomagnetism to Sedimentary Geology*, edited by D. Aissaoui, D. McNeill, and H. Hurley, *SEPM Spec. Publ.*, *49*, 115–128.

Elmore, R. D., S. Banerjee, T. Campbell, and G. Bixler (1999), Paleomagnetic dating of ancient fluid-flow events and paleoplumbing in the Arbuckle Mountains, southern Oklahoma, in *Dating and Duration of Fluid Flow Events and Rock-Fluid Interaction*, edited by J. Parnell, *Geol. Soc. Spec. Publ.*, *144*, 9–25.

Elmore, R. D., J. Kelley, M. Evans, and M. Lewchuk (2001), Remagnetization and orogenic fluids: Testing the hypothesis in the central Appalachians, *Geophys. J. Int.*, *144*, 568–576.

Enkin, R. J., and G. S. Watson (1996), Statistical analysis of palaeomagnetic inclination data, *Geophys. J. Int.*, *126*, 495–504.

Fisher, R. A. (1953), Dispersion on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295–305.

Fruit, D., R. D. Elmore, and S. Halgedahl (1995), Remagnetization of the folded Belden Formation, *J. Geophys. Res.*, *100*, 15,009–15,023.

Gao, G., R. D. Elmore, and L. S. Land (1992), Geochemical constraints on the origin of calcite veins and associated limestone alteration, Ordovician Viola Group, Arbuckle Mountains, Oklahoma, U.S.A., *Chem. Geol.*, *98*, 257–269.

Gill, J. D., R. D. Elmore, and M. H. Engel (2002), Chemical remagnetization and clay diagenesis: Testing the hypothesis in the Cretaceous sedimentary rocks of northwestern Montana, *Phys. Chem. Earth*, *27*, 1131–1139.

Gillett, S. L., and J. W. Geissman (1993), Regional late Paleozoic remagnetization of shallow water limestones in the eastern Great Basin: The usefulness of "micro"-fold tests from late compaction fabrics, in *Applications of Paleomagnetism to Sedimentary Geology*, edited by D. Aissaoui, D. McNeill, and H. Hurley, *SEPM Spec. Publ.*, *49*, 129–147.

Gillett, S. L., and M. E. Taylor (1986), Triassic remagnetization of Lower Paleozoic rocks, Bear River Range, Utah-Idaho: A possible constraint on thermal history, in *1985 Utah Geological Association Guidebook*, edited by G. Kerns and R. Kerns, pp. 249–260, Utah Geol. Assoc., Salt Lake City.

Gillett, S. L., and D. R. Van Alstine (1982), Remagnetization and the tectonic rotation of upper Precambrian and Lower Paleozoic strata from the Desert Range, southern Nevada, *J. Geophys. Res.*, *187*, 10,929–10,953.

Halgedahl, S. L., and R. D. Jarrard (1995), Low-temperature behavior of single-domain through multidomain magnetite, *Earth Planet. Sci. Lett.*, *130*, 127–139.

Heller, P. L., S. S. Bowdler, H. P. Chambers, J. C. Coogan, E. S. Hagen, M. W. Shuster, N. S. Winslow, and T. F. Lawton (1986), Time of initial thrusting in the Sevier orogenic belt, Idaho-Wyoming and Utah, *Geology*, *14*, 388–391.

Hirt, A. M., A. Banin, and A. U. Gehring (1993), Thermal generation of ferromagnetic minerals from iron-enriched smectites, *Geophys. J. Int.*, *115*, 1161–1168.

Huang, W.-L., J. M. Longo, and D. R. Pevear (1993), An experimentally derived kinetic model for smectite to illite conversion and its use as a geothermometer, *Clays Clay Miner.*, *41*, 162–177.

Hudson, J. D. (1977), Stable isotopes and limestone lithification, *J. Geol. Soc. London*, *133*, 637–660.

Huntoon, J. E., P. L. Hansley, and N. D. Naeser (1999), The search for the source rock for the giant Tar Sand Triangle accumulation, southeastern Utah, *AAPG Bull.*, *83*, 467–495.

Jewell, P. W., N. J. Silberling, and K. M. Nichols (2000), Geochemistry of the Mississippian Delle Phosphatic event, eastern Great Basin, U.S.A., *J. Sediment. Res.*, *70*, 1222–1233.

Katz, B., R. D. Elmore, M. Cogoini, M. H. Engel, and S. Ferry (2000), Associations between burial diagenesis of smectite, chemical remagnetization, and magnetite authigenesis in the Vocontian trough, SE France, *J. Geophys. Res.*, *105*, 851–868.

Keith, M. L., and J. N. Weber (1964), Carbon and oxygen isotopic compositions of selected limestone and fossils, *Geochim. Cosmochim. Acta*, *28*, 1787–1816.

Kirschvink, J. L. (1980), The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718.

Lowrie, W. (1990), Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett.*, *17*, 159–162.

McCabe, C., and R. D. Elmore (1989), The occurrence and origin of late Paleozoic remagnetization in the sedimentary rocks of North America, *Rev. Geophys.*, *27*, 471–494.

McCrea, J. M. (1950), On the isotopic chemistry of carbonates and a paleothermometer scale, *J. Chem. Phys.*, *18*, 849–857.

McFadden, P. L., and M. W. McElhinny (1990), Classification of the reversal test in paleomagnetism, *Geophys. J. Int.*, *103*, 725–729.

Melker, M. D., and J. W. Geissman (1997), Paleomagnetism of the Oquirrh Mountains and implications for the Cenozoic structural history of the easternmost Great Basin, in Geology and Ore Deposits of the Qquirrh and Wasatch Mountains, Utah, *Soc. Econ. Geol. Guideb. Ser.*, *29*, 91–112.

Morris, H. T., and T. S. Lovering (1961), Stratigraphy of the east Tintic Mountains, Utah, *U. S. Geol. Surv. Prof. Pap.*, *361*, 145 pp.

Nichols, K. M., and N. J. Silberling (1990), The Delle Phosphatic Member: An anomalous part of the Mississippian (Osagean-Meramecian) shelf sequence of central Utah, *Geology*, *18*, 46–49.

Nichols, K. M., and N. J. Silberling (1993), Upper Devonian to Mississippian strata of the Antler foreland in the Leppy Hills, easternmost northern Nevada, *U.S. Geol. Surv. Bull.*, *1988-G*, G1–G13.

Nichols, K. M., N. J. Silberling, and L. A. McCarley (2002), Regional pattern of Mesozoic structures in the Confusion Range, westernmost central Utah, *Geol. Soc. Am. Abstr. Programs*, *34*, 45.

Nielsen, R. L. (1982), Post-Pennsylvanian structural evolution of southwestern Utah, in *Overthrust Belt of Utah: Symposium and Field Conference, Publ. 10*, pp. 101–115, Utah Geol. Assoc., Salt Lake City.

21562202b, 2004, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2003JB002698 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Oliver, R. (1992), The spots and stains of plate tectonics, *Earth Sci. Rev.*, *32*, 77–106.

Pevear, D. R. (1999), Illite and hydocarbon exploration, *Proc. Natl. Acad. Sci. U.S.A.*, *96*, 3440–3446.

Plaster-Kirk, L. E., R. D. Elmore, M. H. Engel, and S. W. Imbus (1995), Paleomagnetic investigation of organic-rich lacustrine deposits, Middle Old Red Sandstone, Scotland, *Scot. J. Geol.*, *32*, 97–105.

Poole, F. G., and G. E. Claypool (1984), Petroleum source-rock potential and crude-oil correlation in the Great Basin, in *Hydrocarbon Source Rocks of the Greater Rocky Mountain Region*, pp. 179–230, Rocky Mt. Assoc. Geol., Denver, Colo.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop (1975), Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett.*, *28*, 133–143.

Sandberg, C. A., and R. C. Gutschick (1980), Sedimentation and biostratigraphy of Osagean and Meramecian starved basin and foreslope, western United States, in *Paleozoic Paleogeography of the West-Central U.S.*, *Rocky Mt. Paleogeogr. Symp.*, vol. 1, edited by T. D. Fouch and E. R. Magathan, pp. 129–145, Rocky Mt. Sect., SEPM, Denver, Colo.

Sandberg, C. A., and R. C. Gutschick (1984), Distribution, microfauna, and source-rock potential of Mississippian Delle Phosphatic Member of Woodman Formation and equivalents, Utah and adjacent states, in *Hydrocarbon Source Rocks of the Greater Rocky Mountain Region*, pp. 135–178, Rocky Mt. Assoc. Geol., Denver, Colo.

Sandberg, C. A., F. G. Poole, and R. C. Gutschick (1980), Devonian and Mississippian stratigraphy and conodont zonation of the Pilot and Chainman shales, Confusion Range, Utah, in *Paleozoic Paleogeography of the West-Central U.S.*, *Rocky Mt. Paleogeogr. Symp.*, *1*, 71–79.

Silberling, N. J., J. H. Trexler Jr., K. M. Nichols, P. W. Jewell, and R. A. Crosbie (1997), Overview of Mississippian depositional and paleotectonic history of the Antler foreland, eastern Nevada and western Utah, *Brigham Young Univ. Geol. Stud.*, *42*, 161–197.

Stamatakos, J., D. A. Ferril, and K. H. Spivey (1998), Paleomagnetic constraints on the tectonic evolution of Bare Mountain, Nevada, *Geol. Soc. Am. Bull.*, *110*, 1530–1546.

Stueber, A. M., P. Pushkar, and E. A. Hetherington (1984), A strontium isotopic study of Smackover brines and associated solids, southern Arkansas, *Geochim. Cosmochim. Acta*, *48*, 1637–1649.

Swart, P. K., S. J. Burns, and J. J. Leder (1991), Fractionation of the stable isotopes of oxygen and carbon in carbon dioxide during the reaction of calcite with phosphoric acid as a function of temperature and technique, *Chem. Geol.*, *86*, 89–96.

Symons, D. T. A., and M. T. Cioppa (2000), Crossover plots: A useful method for plotting SIRM data in paleomagnetism, *Geophys. Res. Lett.*, *27*, 1779–1782.

Van der Voo, R. (1993), *Paleomagnetism of the Atlantic, Tethys, and Iapetus Oceans*, Cambridge Univ. Press, New York.

Watson, G. S., and R. J. Enkin (1993), The fold test in paleomagnetism as a parameter estimation problem, *Geophys. Res. Lett.*, *20*, 2135–2137.

Weil, A. B., and R. Van der Voo (2002), Insights into the mechanism for orogen-related carbonate remagnetization from growth of authigenic Fe-oxide: A scanning electron microscopy and rock magnetic study of Devonian carbonates from northern Spain, *J. Geophys. Res.*, *107*(B4), 2063, doi:10.1029/2001JB000200.

Zijderveld, J. D. A. (1967), A.C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D. E. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier Sci., New York.

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